

The Story of Birds: A New History From Their Dinosaur Origins

Extract from Steve Brusatte's new book · By Stephen Brusatte, University of Edinburgh · Published April 28, 2026

Source: theconversation.com/...-extract-of-steve-brusattes-new-book-281562

I will never forget my first dinosaur wing. I was a college student, on my first international expedition, preparing to venture into the mountains of Tibet in search of Jurassic dinosaurs.

Our team assembled in Beijing, and as we rushed through the galleries and storehouses of the Institute of Vertebrate Paleontology and Paleoanthropology, I stole a fleeting glance, from across the room. A skeleton of the little carnivore *Microraptor*, its long arms unfurled, adorned with feathers forming a broad sheet. The wings sparkled in the low light; I was mesmerised. And then we were hustled along.

Nearly a decade later, I got to spend quality time with a dinosaur wing. My friend Junchang Lü, one of China's leading dinosaur hunters, had gotten word that a farmer in Liaoning had stumbled upon something remarkable while harvesting his crops. It was a fossil coelurosaur (a variety of two-legged dinosaurs that includes modern birds), said to be swathed in many types of feathers. A few grainy photos confirmed those details, but little more.

A museum in the city of Jinzhou had procured the specimen and invited Junchang to study it. Because I had analysed many coelurosaurs for my PhD thesis, Junchang asked for my help. We met one cold November morning at Beijing's central railway station and boarded an eastbound train. We didn't know what would be waiting for us when we disembarked.

Two black SUVs, as it turned out. Junchang and I were whisked inside, and we sped through the streets of Jinzhou like we were in a presidential motorcade. When we arrived at the museum, we were led through a dim hallway and into a side room, where a big slab of gray rock balanced on a table. Everyone paused. A strange tension filled the air. After a few whispered words of Mandarin, Junchang turned to me and motioned us forward.

In front of us was a dinosaur skeleton the size of a large dog. It was obviously a dinosaur, as it had the reinforced pelvis, with extra connections between the backbone and hips, that all

dinosaurs inherited from their Triassic ancestors.

Beyond that, I could tell it was a dromaeosaurid – a member of the same "raptor" group as Ostrom's *Deinonychus*, Jurassic Park's *Velociraptor*, and the *Microraptor* I'd seen years earlier in Beijing – because it had the signature sickle claws on its toes. And there were indeed feathers all over its body, which a couple of decades earlier would have seemed outlandish, but by now was no surprise to us.

What was astonishing, though, were its arms. Quite simply, the arms were wings. And the individual feathers composing the wings were preserved in sublime condition. We could clearly see a line of ten fan-shaped feathers attaching to the hand, each one longer than the humerus bone of the upper arm. Making a continuous series with them, another 20 feathers followed behind, affixed to the ulna, the bow-shaped long bone of the forearm that forms the wrist where it meets the hand, and the elbow joint at its back end.

Partially covering all of these feathers, close to where they attached to the arm bones, was a blanket of about 30 additional quills. These overlain feathers were smaller than the ones attaching to the hand and forearm, so they formed a dense covering close to the arm bones, but did not extend all the way across the wing.

This was the wing of a bird. Its overall construction was exactly that of a sparrow or an eagle. If I had just seen this wing and not the raptor dinosaur it was connected to, I would probably think that it belonged to some large bird.

The feathers attached to the hand are the primaries. They are the longest and narrowest of the wing feathers, can rotate individually relative to each other, and form much of the front and side of the wing when it is unfolded. The feathers attached to the ulna are the secondaries. They are more crowded together than the primaries, and therefore make a more coherent sheet, which forms much of the back edge of the wing.

The secondaries often join to the bone via ligaments, which attach to a series of bumps along the ulna called quill knobs. And finally, the feathers that covered the primaries and secondaries are the coverts. They help protect the primaries and secondaries, and give the wing extra integrity.

We named this winged raptor *Zhenyuanlong*: Mr. Zhenyuan's dragon, in honour of the museum director who secured it from the farmer. Because its wing so closely matches that of a modern bird, and because today's birds use their wings to fly, you would probably assume that *Zhenyuanlong* was soaring over the Cretaceous volcanoes of China.

But it didn't. It couldn't. It was not capable of doing that special thing that birds can do: actively flap its wings to generate enough of those two key aerial forces: lift to get airborne and thrust to move forward through the air. This is what is called powered flight, as opposed to passive aerial manoeuvres such as gliding.

Zhenyuanlong's body was too big, and its wings too small in proportion: if it tried to flap it wouldn't have been able to stay airborne. That's not a guess; it's the laws of physics. There is a well-known relationship between body size and wing size in birds today that are capable of powered flight. They can have approximately 2.5 grams of body weight for every square centimetre of wing area and, at least theoretically, be able to support their body in the air. Any more weight, and flapping doesn't work.

Zhenyuanlong was nowhere close to this cutoff; it would have needed to lose about half its weight in order for its wings to work as flappers. Other aspects of its skeleton corroborate this: its arms are quite short, and it lacks the large breastbone (sternum) that anchors the bulging flight muscles of today's birds. Maybe *Zhenyuanlong* could have glided a short distance, but that would have been the extent of its aerial adventures.

Sexual Selection

What we see in *Zhenyuanlong* is part of a wider trend. Many coelurosaurs had wings of quill feathers, but their wings were too small to power their bodies through the air. If you look at trends in feather evolution across the family tree of dinosaurs, the first species with wings were oviraptorosaurs like *Caudipteryx* and *Big Mama*, animals the size of sheep with wings no bigger than a dinner plate.

There is provocative evidence from fossil quill impressions on the ulna bones of even more primitive coelurosaurs that wings might have even first evolved in horse-size dinosaurs. Those wings would likely have been about the size of laptop screens. There's no way any of these dinosaurs could fly with such wings – at least in an active, flapping way.

This leads to a startling realisation: dinosaur wings did not evolve for flight. Like feathers themselves, wings evolved for another reason, and were later repurposed as airfoils. It's a conundrum. Why else would such a large and complicated structure as a wing develop in the first place?

Looking at the too-tiny-to-flap wings of *Zhenyuanlong* got me thinking about a famous quote from Charles Darwin. "The sight of a ... peacock's tail, whenever I gaze at it, makes me sick!" he

wrote to his botanist friend Asa Gray in April 1860, less than six months after he published *On the Origin of Species*.

Although Darwin was notoriously prone to stomach pains and other ailments, this particular ache was metaphorical. In the flamboyant train of the peacock – built from huge feathers longer than the bird's body – Darwin saw an outrageous structure whose origins he could not comprehend. It wasn't used for flying, even though it was made of big feathers. It wasn't helpful in finding food or escaping predators.

Try as he might, he could not envision it evolving through natural selection, the mechanism for change over time that he had just articulated in his new book, in which variations that confer advantages are favoured through survival of the fittest.

Perhaps, Darwin surmised, beautiful feathers and other gaudy structures develop not because they help their bearers survive the perils of droughts and predators, but because they promote reproduction. He called this theory "sexual selection", to distinguish it from natural selection.

In 1871, in *The Descent of Man, and Selection in Relation to Sex*, Darwin explained how sexually selected traits like fanciful crests, feathers and colour patterns improve an individual's ability to acquire mates, by making them either more attractive to potential partners or better able to compete with rivals in the mating game through displays of dominance and intimidation.

Might sexual selection explain why some coelurosaurs turned their simple fuzzy feathers into wings made of quills. It's hard to prove definitively, but I think several lines of evidence support the case.

First is the fact that many modern birds like peacocks use their feathers – and even embellish them into comically large billboard-like structures – to attract mates and intimidate rivals. Second is the fact that sexual selection was operating in some of the first true birds flying overhead of their ground-bound coelurosaur cousins.

A primitive bird from China, called *Confuciusornis*, is known from hundreds of fossils, half of which have ridiculously long ribbon feathers streaming off their tails, and half of which don't. These feathers are too skinny to provide any lift or thrust during flight, they're seen in only half the population, and, crucially, they're not present in those individuals that have medullary bone, the unique tissue that female birds use to mine calcium to shell their eggs.

Clearly these tawdry ribbon feathers are the stuff of males, and their only plausible use could be in display. Sexual selection, therefore, was happening in birds from the very beginning of

powered flight, so it's likely it was shaping their coelurosaur ancestors too. The most convincing evidence, however, comes from the coelurosaur fossils themselves.

Dinosaurs in Colour

Many of the dinosaur books I read as a child in the late 1980s and 1990s would include a defeatist statement: we'll never know what colours dinosaurs were. Those books were wrong.

In the late 2000s a tall, bearded Danish PhD student named Jakob Vinther was looking at fossils under high-powered scanning electron microscopes and noticed a peculiar detail. Many of them – including dinosaur feathers – preserved a variety of small, bubble-like structures. They looked exactly like the melanosomes of modern-day animals. These are the little vessels that hold pigment, the chemicals that confer colour. We know that sausage-shaped melanosomes impart a black hue, meatball-shaped ones a rusty red, and so on.

By measuring the size and shape of the fossil melanosomes, and comparing them to melanosomes in today's animals, Jakob could predict the colours of dinosaur feathers. It was an astounding revelation.

Before long, Jakob and his colleagues showed that coelurosaurs boasted a brilliant array of plumage. Some feathers were black, others white, gray, or ginger. One little winged coelurosaur called *Anchiornis* was decked out in a fancy coat, as if going to a Jurassic cocktail party. Its face was speckled black and red, a ginger mohawk erupted from its head and neck, and its wing feathers were white across most of their lengths but black at their tips, which when the various primaries and secondaries and coverts were layered together, produced a wing of alternating white and black stripes, like the hide of a zebra.

Later Jakob got me in on the action, and trained one of my undergraduates, Angus Croudace, in deciphering the colours of another coelurosaur, a raptor called *Wulong*, which had a drab grey body but black wings that sparkled in the sun with the iridescent sheen of a crow.

Such excessive flaunting of colour and texture could have served only one purpose: display. These coelurosaur ancestors and cousins of birds were under the spell of sexual selection, and some of the feathers forming the first dinosaur wings – which were useless as flapping airfoils – were being used as ornaments.

Wings, therefore, might have first evolved in dinosaurs as advertising billboards sticking off of the arms. That was probably what many of these winged coelurosaurs, like *Zhenyuanlong*, were using their wings for.

Lift Off

Then the billboards took on a new function, and became airfoils. It would have been pure happenstance. Remember that these coelurosaurs were getting smaller over time, and that there is also a tendency for display structures to become ever more fanciful in order to stay ahead in the mating race.

So while coelurosaurs like *Zhenyuanlong* were too big to fly with their small wings, you can imagine that if their bodies got a bit smaller, and their wings a bit bigger in order to look more attractive to mates or scary to rivals, a tipping point would be reached.

By the laws of physics, the billboards would now be broad enough, relative to the smaller size of the body, that if the coelurosaur moved them around, they could produce a little lift, a little thrust, and that dinosaur could start flickering about in the air.

These first attempts at flight would have been awkward, and the billboard wings might have initially been used more as brakes or stabilisers, to help in leaping and turning. But now, a threshold had been crossed, and Darwin's classic natural selection could start fine-tuning these flapping dinosaurs into ever-better aerialists.

It's a story that makes intuitive sense. A nice, orderly progression from winged coelurosaurs to flying birds. But like many things in nature, the reality was probably much more complicated – and much more interesting.

The more feathered dinosaurs we find, the more we realise that not all coelurosaurs had the same types of wings. Some, like *Zhenyuanlong*, look similar to today's birds, with wings on the arms made from layers of primaries, secondaries and coverts. But then there's *Microraptor* – the little dromaeosaurid that hypnotised me as a college student in Beijing – which not only had wings on its arms but also on its legs, and its tail too.

In fact, such hind limb wings are present in many other feathered coelurosaurs, which is odd, because although various birds today have feathers on their legs, none has anything approaching a broad, sheetlike wing. And weirdest of all is tiny *Yi qi*, a fluffy coelurosaur that would fit in the palm of your hand, which had feathers on its body but a wing made of something else entirely: skin, which stretched between its fingers and a strange rod-like bone in its wrist. It was a dinosaur that looked like a bat.

Some of these weirdly winged dinosaurs could surely glide, and a few probably were capable of powered flight. *Microraptor* is the prime example. My colleagues and I have done the calculations, and the wings of *Microraptor* were more than big enough to support its crow-size

body in the air. On top of that, its feathers show many harbingers of flapping flight in modern birds.

Microraptor's wing feathers are asymmetrical, with a much narrower leading vane in front of the shaft and a wider trailing vane. This is a telltale aerodynamic signal: It allows the overlapping feathers to form the cambered shape that a wing needs in order to function as an airfoil that can generate lift, the same way an airplane wing is curved at the top.

One stunning *Microraptor* fossil shows its wing feathers in mid-molt, and like flying birds today, it lost old feathers and grew new ones in a sequential pattern rather than all at once, probably so it could maintain enough of an airfoil to fly and not have to stay grounded for days at a time while it replenished its plumage. And finally, perhaps most convincing of all, scientists have built physical models of *Microraptor*, put them into wind tunnels, and observed how the wings are able to generate lift to keep the model airborne.

These various dinosaur wing shapes are indicative of distinct ways of gliding and flying, which, to me, implies multiple independent origins of flight among dinosaurs. We can understand it with an analogy to human aeronautics. A hot air balloon and a Boeing 737 can both get airborne, but they do so in much different ways, and look nothing alike.

No single engineer turned a hot air balloon into a Boeing 737; instead, they developed separately as different people, at different times, tinkered with ways to get into the air. But, those engineers had a common understanding of flight mechanics and properties like lift and thrust, so they built their flying machines out of the same general knowledge base.

Evolution seems to have done something similar with dinosaurs. There was a zone on the family tree of small coelurosaurs whose ancestors had already developed feathers for insulation, and elaborated them into wings for display.

Here and there, sexual selection and natural selection could make little tweaks to this common blueprint – a small decrease in body weight here, a slightly bigger advertising billboard wing there – and different coelurosaurs would be able to get airborne. Each followed its own idiosyncratic route into the skies, some probably soaring upward from the ground and others parachuting downward from the trees, but they all emerged out of this frenzy of experimentation.

If you were back in the Jurassic or Cretaceous, trying to avoid the footfalls of a *Brontosaurus* or the crushing bite of a *T.rex*, the skies would have been aflutter with a prehistoric aviary of gliding and flapping dinosaurs. It's hard to know exactly how many coelurosaurs

independently invented flight – it might have been just a few groups, it might have been dozens.

Some were probably short-lived; others might have persisted for millions of years, particularly those raptors with arm and leg wings. All bar one, however, met the same ultimate fate: extinction. That one survivor was the group that led to modern birds, which actively flap their arm wings made of primaries, secondaries and coverts.

It would be like if the entire history of human aeronautics – every hang glider, weather balloon, prop plane, crop duster, helicopter, rocket, jumbo jet – was wiped away, except for, say, space shuttles.

Copyright © 2026 by Stephen (Steve) Brusatte. Published by Mariner Books, an imprint of HarperCollins Publishers. Reprinted with permission via The Conversation. The US edition publishes April 28, 2026; UK edition June 11, 2026.